

Spectrum of Dissent Coding Protocol

May 5, 2023

Section 1: Inclusion/Exclusion Procedure: This section speaks to whether the article will be included in the final sample for analysis.

- A. Is the article significantly relevant? Relevance is defined as containing species mention and being of suitable coverage – Event announcements, advertisements, etc. were excluded for non-significance, as they would not likely include much information pertinent to that being collected.
 - a. Code as Yes or No
- B. Is the article primarily* about the species of interest? Primarily* = Species of Interest is mentioned in headline and/or the article is 75% or more about the species of interest
 - a. Yes – Code as PRIMARY
 - b. No – Code as FAIL* (FAIL* includes duplicate materials)
- C. Does the article reference species specifically in Hawai'i?
 - a. Yes – Code as HI
 - b. No – Code as FAIL
- D. Is the species discussed in captivity/zoo?
 - a. No – Code as Wild
 - b. Yes – Code as FAIL
- E. Does the article contain any FAIL codes?
 - a. Yes – EXCLUDE from final corpus.
 - b. No – Article passes all inclusion criteria. INCLUDE in final corpus.

Section 2: General Article Information: These variables speak to article metadata/general information.

- A. Article Number - Article numbers are assigned after-the fact; 1-445
- B. Article Date - Date should be coded as follows: MM/DD/YYYY
- C. Article Title – Coded verbatim
- D. Article Page Number – coded verbatim
- E. Name of Publication - coded verbatim; to include:
 - a. Hawaii Tribune-Herald
 - b. Honolulu Advertiser
 - c. Honolulu Star-Advertiser
 - d. Honolulu Star-Bulletin
 - e. Garden Isle
 - f. West Hawaii Today
- F. Author – Coded verbatim
- G. Article Type – Coded as follows:
 - a. News Article – covers interviews, policy, events, emerging information, etc. and is primarily written as narrative by a reporter. (1)
 - b. Column/Feature Article – a piece written from the view of the author as part of a regular series. Ex: Gardening Tips, Humor articles, etc. (2)
 - c. Editorial – shares opinion of the news organization (3) ***Combined with Opinion (d)
 - d. Letters/Commentary/Opinion – opinion-based commentary or responses to news coverage; written by members of the public. (4)

- H. Valence – What is the primary depiction of the species? Code as Negative, Positive, or Neutral.
 - a. (1) Negative = Species is primarily described negatively (at least one or more instances).
 - b. (2) Positive = Species is primarily described positively (at least one or more instances).
 - c. (3) Neutral = Species is described both positively and negatively in about equal amounts.

Section 3: Invasion History Information

This section is designed to identify which impacts and risks authors choose to highlight, as well as language which is most used to discuss invasive species.

- A. Language – Which words are used to describe the species? Code with word used. The list will be adapted as needed.
 - a. None (0)
 - b. Invasive (1)
 - c. Pest (2)
 - d. Alien (3)
 - e. Multiple (4)
 - f. Introduced (5)
 - g. Nonnative (6)
- B. Invasional Meltdown – Does the article suggest that the species of interest will lead to subsequent establishment of other invasive species?
 - a. Code Y/N and quote
- C. Impact – Does the article reference impacts related to the species establishment, spread, and/or eradication? Note: Impact should be distinguished from risk. Impact is an observed effect.
 - a. Code Y/N and quote
- D. Risk – Does the article reference risks associated with the species establishment, spread, and/or eradication? Note: Risk should be distinguished from Impact. Risk is an anticipated or potential effect.
 - a. Code Y/N and quote

Section 4: Dissent

This section serves to identify if dissent is present regarding the species of interest. Dissent is identifiable by the presence of disagreement, disbelief, and/or questioning surrounding species of interest.

- A. Is dissent evident?
 - a. Code Y/N and quote

Section 5: Timeline

A timeline variable was included to assist with reconstruction of the invasion process, and better identify when dissent is most prevalent within the invasion process. Similarly, descriptive words were coded to gain a clear picture of how the species was being painted.

- A. Are key events included/described? Examples include, but not limited to:
 - a. New science published on species.
 - b. New eradication method deployed for species.
 - c. New law passed related to species.

- d. County declared fully eradicated.
- B. Which words were used to describe the species of interest?
- Code them as worded. Example, if species is described as loud, code “loud”.

Supplementary Note: The following variables were dropped from this study –

Variable	Definition		Underlying Problem
Uncertainty	Utilize Szarvas et al.’s uncertainty cues to identify semantic uncertainty and categorize according to Latombe et al.	(Latombe et al., 2019; Szarvas et al., 2012)	It was not always clear whether semantic uncertainty was introduced by the author or in relation to science.
Impact Valence	Identifies the overall valence of impacts described (i.e., positive/negative)	(Golebie et al., 2022)	Valence was not always overwhelmingly positive or negative – prompting a grey space.
Overall Topical Subject	Identifies the overall topical framing (e.g., eradication, impact, etc.) of the media article	(Muter et al., 2012)	The title did not always aptly reflect the overall topical balance of the article; articles often covered multiple aspects (e.g. spread, impact, eradication)
Quoted Individuals	Sources quoted are known to influence public perception. This variable identified speakers and their role (e.g., scientist, policymaker, retail worker, etc.)	(Jacobson et al., 2012)	Often times many individuals were quoted. The boundaries of their work were not always clear. Ex: X individual from Coqui Frog Working Group – are they a academic scientist? Government worker? A scientist in government?

- Golebie, E. J., Riper, C. J. v., Arlinghaus, R., Jang, S., Kochalski, S., Stedman, R., Lu, Y., Olden, J. D., & Suski, C. (2022). Words matter a systematic review of communication in non-native aquatic species literature. *Neobiota*, 74, 1-28. <https://doi.org/10.3897/neobiota.74.79942>
- Jacobson, S. K., Langin, C., Carlton, J. S., & Kaid, L. L. (2012). Content Analysis of Newspaper Coverage of the Florida Panther. *Conservation Biology*, 26(1), 171-179. <https://doi.org/10.1111/j.1523-1739.2011.01750.x>
- Latombe, G., Canavan, S., Hirsch, H., Hui, C., Kumschick, S., Nsikani, M. M., Potgieter, L. J., Robinson, T. B., Saul, W.-C., Turner, S. C., Wilson, J. R. U., Yannelli, F. A., & Richardson, D. M. (2019). A four-component classification of uncertainties in biological invasions: implications for management. *Ecosphere*, 10(4), 1-25. <https://doi.org/10.1002/ecs2.2669>
- Muter, B. A., Gore, M. L., Gledhill, K. S., Lamont, C., & Huveneers, C. (2012). Australian and U.S. News Media Portrayal of Sharks and Their Conservation. *Conservation Biology*, 27(1), 187–196. <https://doi.org/10.1111/j.1523-1739.2012.01952.x>
- Szarvas, G., Vincze, V., Farkas, R., Mora, G., & Gurevych, I. (2012). Cross-Genre and Cross-Domain Detection of Semantic Uncertainty. *Association for Computational Linguistics*, 38(2).